

Oil and gas well development and coccidioidomycosis risk in Kern County, CA, USA: a case-crossover study



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Summary

Background Coccidioidomycosis is an emerging fungal disease caused by inhaling *Coccidioides* spp spores. As spores reside in soil, activities that disturb soil and generate dust can aerosolise and transport the pathogen. The oil and gas industry has been extensively developed in some regions that are endemic for *Coccidioides* spp and has been associated with dust emissions. Although several adverse health outcomes have previously been associated with oil and gas development, its impact on coccidioidomycosis risk has not been investigated. We aimed to estimate the association between exposure to oil and gas well development (ie, wells in preproduction) and risk of coccidioidomycosis among residents living near new wells.

Methods In this case-crossover study, we obtained information on reported coccidioidomycosis cases and oil and gas well development between 2007 and 2022 in Kern County, CA, USA. We then compared exposure to preproduction wells within 5 km of each individual's place of residence during both hazard (ie, the 49–139 days before case onset) and control periods using conditional logistic regression.

Findings During the study period, 658 108 (72.4%) of 909 282 of Kern County residents lived within 5 km of at least one preproduction well, and 116 020 (12.8%) lived within 5 km of 23 or more preproduction wells within a single 90-day period. We estimated that the odds of coccidioidomycosis incidence were 12.5% (95% CI 5.8–19.6) higher in the 90 days following exposure to at least one preproduction well within 5 km of an individual's place of residence and that the odds of infection increased by 0.7% (0.4–0.9) for each additional preproduction well developed within this distance.

Interpretation These findings support a previously-unrecognised association between the development of oil and gas wells and transmission of coccidioidomycosis, potentially driven by dust generation. Given the prevalence of oil and gas development in the study region, its impact on coccidioidomycosis incidence might be large.

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Introduction

Coccidioidomycosis (Valley fever) is an emerging infectious disease endemic in the southwest region of the USA. More than 200 000 symptomatic cases are estimated to occur in the USA each year.¹ Disease manifestations range from a self-limiting respiratory infection to a disseminated disease,^{2,3} which can be fatal. Infection results from inhaling spores of *Coccidioides* fungi, which reside in soil and can become airborne through disturbances such as high wind and dust events, earthquakes, construction, and farming.^{4–6} Infectious spores (arthroconidia) can travel up to several miles in dust and can be inhaled directly or settle and become resuspended in downwind locations.⁷ Although increased exposure to ambient fine mineral dust has been associated with increased incidence of coccidioidomycosis in the state of California,⁸ the specific sources of dust emissions remain largely unidentified.

Oil and gas development has previously been associated with emissions of mineral dust and several other health-damaging air pollutants, including other particulate

matter, ozone, and non-methane volatile organic compounds.^{9,10} The construction of new wells might be particularly likely to generate dust as the development process involves many soil-disturbing steps, including transporting heavy equipment, clearing and levelling the site, constructing the well pad, digging water and waste pits, and drilling a bore hole—steps collectively termed preproduction.^{9,11,12} Although several adverse health outcomes have been associated with oil and gas well development—including preterm birth, asthma, reduced lung function, paediatric haematological cancer, COVID-19 severity, and all-cause mortality^{13,14}—its impact on coccidioidomycosis risk has not been investigated.

We aimed to investigate the association between coccidioidomycosis incidence and oil and gas development in Kern County, CA—one of the top seven oil-producing counties in the USA¹⁵ and consistently among the top three counties for coccidioidomycosis incidence in the USA (figure 1).¹⁶ We also aimed to investigate whether any association between exposure and coccidioidomycosis

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Research in context

Evidence before this study

Coccidioidomycosis is an emerging fungal disease caused by inhalation of airborne *Coccidioides* spores. As these spores live in the soil, activities that disturb the soil—such as construction, farming, earthquakes, and dust events—have previously been associated with increased transmission risk. However, the full range of activities that facilitate pathogen transmission remains poorly characterised. We searched PubMed from database inception to Feb 11, 2026, for articles published in English with the search terms “oil well” OR “gas well” OR “oil and gas construction” OR “oil and gas development” AND “infectious disease” OR “transmission” OR “risk”, as well as their common textual variants. We identified 35 relevant studies that investigated the associations between oil and gas development and adverse health outcomes, including preterm birth, low birthweight, cancer diagnoses, upper respiratory symptoms, and all-cause mortality. Only one previous study investigated infectious disease outcomes and found that high levels of exposure to oil and gas production were associated with moderately elevated COVID-19 severity. We did not identify any previous studies that had investigated the impact of oil and gas development on the risk of coccidioidomycosis or any other environmental pathogens.

Added value of this study

In this study, we investigated whether oil and gas development contributes to coccidioidomycosis risk in an endemic area, given

that it involves several soil-disturbing processes (eg, clearing of the site, levelling of the ground, and movement of heavy equipment). Using a case-crossover study design focused on Kern County, CA—one of the top seven oil-producing counties in the USA—we found that the development of new oil and gas wells was associated with elevated coccidioidomycosis risk among individuals living within 5 km of the wells. Associations were strongest when the well was developed during fall or summer, when dry soil can be most readily aerosolised. Furthermore, we found that approximately 658 108 (72.4%) of 909 282 of Kern County residents lived near at least one well developed over the study period (2007–22), and 116 020 (12.8%) lived near at least 23 wells developed within a single 90-day period. Considering the high level of exposure, oil and gas development might have a large population-level impact on coccidioidomycosis incidence in this region. This study identifies a previously-unrecognised adverse health outcome of oil and gas development and adds to a growing body of evidence on the harmful health impacts of oil and gas development.

Implications of all the available evidence

Our study adds to a growing body of evidence of harmful health impacts of oil and gas development by identifying an association with an emerging fungal disease. As no vaccines are available for coccidioidomycosis and effective antifungal drugs are few, identifying and mitigating environmental exposures to fungal spores is crucial for protecting public health.

incidence was modified by patient sociodemographics (ie, age, race or ethnicity, and sex), prevailing wind directionality, or season of case onset.

Methods

Study period and region characteristics

We restricted our analysis to Kern County, CA, USA. This is the county with the highest incidence of coccidioidomycosis in the USA (323.2 cases per 100 000 in 2018)¹⁶ and with the most extensive oil and gas development in the state (figure 1).¹⁵ We defined the study period from Jan 1, 2007, to Dec 31, 2022, the range for which data on both coccidioidomycosis cases and oil and gas well construction activities were available.

Data sources

We obtained information on all coccidioidomycosis cases reported in the county from the California Department of Public Health (appendix p 2). Since 2019, cases have been classified on the basis of a positive *Coccidioides* laboratory test result. Before 2019, classification required both laboratory and clinical criteria to be met.¹⁷ The date of case onset was determined on the basis of patient interviews or the earliest recorded clinical or laboratory date. Thus, the estimated date of onset reflected lags between spore inhalation, symptom onset, health-care seeking, and

diagnosis. All case data were de-identified before analysis. Ethics approval for the study was obtained from the Committee for Protection of Human Subjects of the California Health and Human Services Agency (CHHS; protocol number 17-05-2993). Approval by the University of California, Berkeley, was provided on the basis of the CHHS approval.

We obtained information on all oil and gas wells constructed in California from the California Geologic Energy Management Division. These data included the geographical coordinates of each well (figure 2), spud date (ie, the official date when a drill bit first breaks the surface of the ground), completion date, and well status (appendix pp 2, 7). We included only non-cancelled oil and gas wells with an available spud date, completion date, or both during the study period (2007–22) that were located within 5 km of at least one reported case within the study region—a threshold chosen on the basis of findings from a previous work finding that newly constructed oil and gas wells were associated with elevated PM_{2.5} concentrations within this distance (appendix p 2).¹⁰

Exposure definition

We focused on exposures to wells in preproduction—the period between preparing for spudding and the well completion date—to capture potential dust and spore

See Online for appendix

exposures from all the steps used to establish and prepare wells for production (appendix pp 7–8).^{10,12,18} Specifically, we defined the preproduction period for each well as starting one week before the spud date and ending one week after the completion date to account for the full range of oil pad activities that might have generated dust, following a previous study (appendix pp 2, 7).¹⁸ We classified exposures using three metrics: a binary metric (ie, the presence or absence of at least one preproduction well), a continuous metric (ie, the number of preproduction wells), and a quartile metric (ie, whether the number of preproduction wells fell within the first, second, third, or fourth quartiles). For the binary and continuous exposures, we estimated exposure within buffers of varying distances between the places of residence of individuals with reported coccidiomycosis and well sites, ranging from less than 1 km to 5 km (figure 3A). For the quartile exposure, we only estimated exposures occurring between 0 km and 5 km from places of residence. As the number of unexposed cases were substantial, we calculated quartiles using only non-zero exposures and used unexposed cases as the reference group in our statistical models.

Study design

We used a case-crossover approach to examine the association between exposure to preproduction wells and coccidiomycosis incidence. This design enabled us to isolate the impacts of exposure to preproduction wells before case onset while controlling for individual characteristics, such as race, sex, and occupation, which might be associated with exposure, disease, or both, but are mostly time-invariant.^{6,19} Inferences were obtained by comparing the distribution of exposures during an individual's hazard period (ie, before the infection outcome occurs) and a corresponding control period (figure 3B). Accordingly, each individual served as their own control; thus, confounders that were largely time-invariant (eg, sex and race) were inherently adjusted for (appendix pp 3–4).¹⁹

We defined the hazard period as the 49–139 days before case onset (figure 3B). A previous study showed that the associations between dust exposure and reported coccidiomycosis incidence are strongest 1–3 months before case onset, reflecting delays between exposure, symptom onset, and diagnosis.⁸ These lags can vary substantially among individuals, with the interval between exposure and symptom onset estimated at 7–21 days,²⁰ and the interval between symptom onset and diagnosis ranging from less than 10–180 days (median 55 days).³ Thus, we specified a 90-day hazard window beginning 49 days before case onset to approximate the 1–3-month period while allowing for individual-level variation and longer delays in incubation, symptom presentation, health-care seeking, and diagnosis. To ensure our results were robust to hazard period specification, we repeated the analyses under alternative specifications (appendix pp 2–3, 8).

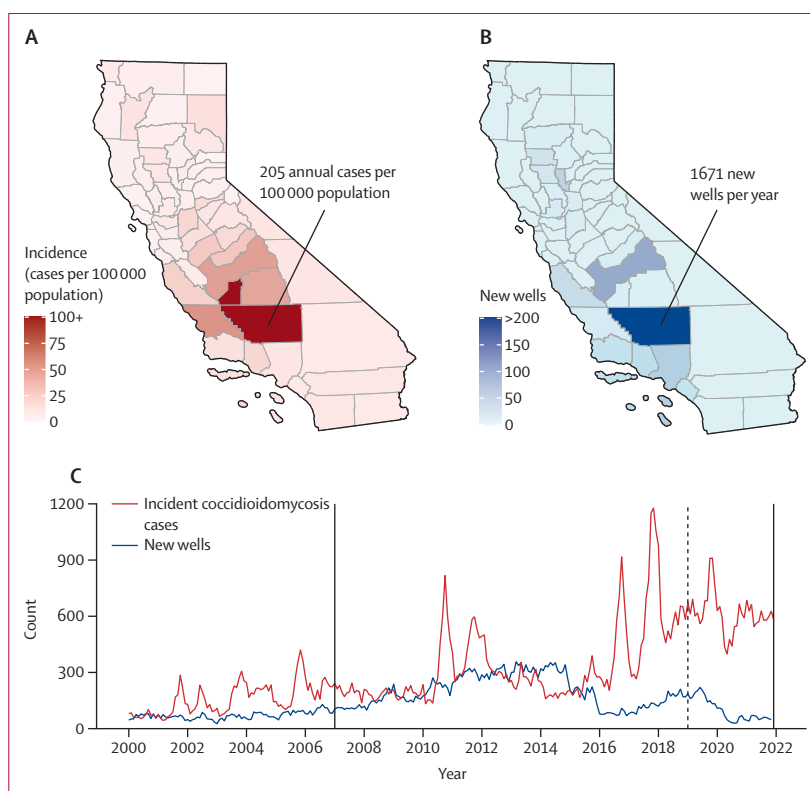


Figure 1: Spatial and temporal trends of coccidiomycosis incidence and oil and gas development (A) Mean annual county-level coccidiomycosis incidence (cases per 100 000 population) in California counties between 2007 and 2022. (B) Mean annual counts of newly constructed wells in California counties between 2007 and 2022. (C) Monthly counts of newly constructed wells (blue) and cases (red) in Kern County, California between 2000 and 2022. The solid vertical lines at 2007 and 2022 delineate the study period. The dashed vertical line at 2019 denotes the year in which the coccidiomycosis case definition changed.

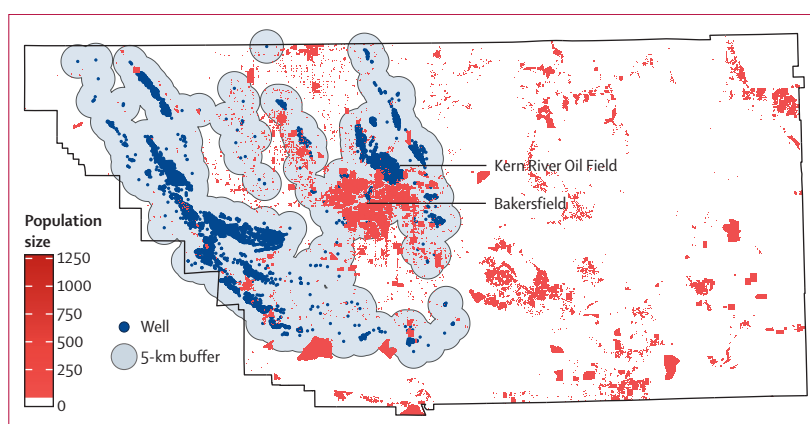


Figure 2: Locations of oil and gas wells constructed during the study period, shown with a 5-km buffer, alongside population density estimates across Kern County, CA, USA

Population estimates were obtained in 100-m grid cells for 2020 from CA-POP, a publicly available dataset of gridded population estimates. Non-coloured grid cells denote unpopulated regions. Bakersfield (the population centre of the county) and the Kern River Oil Field (one of the largest and most dense oil fields in the state) are labelled.

For each patient, we selected a control period using a time-stratified, semisymmetric, bidirectional design. The control period encompassed the same days of the year as the hazard period (to control for seasonality in

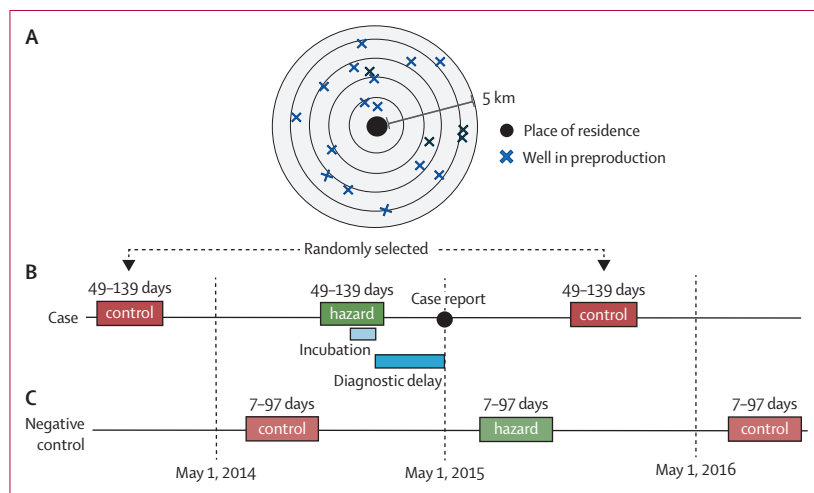


Figure 3: Visual representation of our use of the case-crossover design

(A) Schematic representing our calculation of exposure to oil and gas wells in preproduction during hazard and control periods (ie, by summing the number of preproduction wells within buffers of varying distances up to 5 km from an individual's residence). (B) The case-crossover design, which defined the true hazard period as 49–139 days before the case report and the control period as the same date range in either the following or preceding year, based on a random draw. Sensitivity analyses were conducted for alternative specifications of the hazard period. Incubation denotes the interval between exposure and symptom onset, previously estimated as 7–21 days. Diagnostic delay denotes the interval between symptom onset and diagnosis, previously estimated as a median of 55 days. (C) Negative control, in which the hazard period is defined as 7–97 days after case onset, and control periods were the same date range but occurring in either the following or preceding year.

transmission) but occurred in either the calendar year before or the calendar year after the hazard period (to control for any long-term secular trends; figure 3B). We used conditional logistic regression to regress case occurrence against exposure during the hazard and control periods, using the individual as the stratum. We constructed separate models (appendix pp 3–4) for each exposure metric (ie, binary, continuous, and quartile) and separately, for each of the five exposure distances (ie, 0–1 km, 0–2 km, 0–3 km, 0–4 km, and 0–5 km buffers from places of residence).

To further reduce any residual confounding from any unmeasured, time-varying factors, we included a negative control exposure in our model, following a previous study.²¹ We defined the negative control exposure as a hazard period that occurred in the 7–97 days following a case report (figure 3C). This hazard period is an ideal negative control as the construction of new wells occurring after the estimated disease onset is expected to be associated with the true exposure (as some locations might experience more frequent oil and gas well construction). However, exposures occurring after infection are conditionally independent of disease onset in the absence of confounding, model misspecification, and measurement error.²¹ We used the same design for selecting control periods for the negative controls, using the direction determined by the random draw from the primary exposure (appendix pp 4, 6).

Statistical analysis

We examined the robustness of our results under alternative specifications of the hazard period, exposure distance,

and case definition. Specifically, we repeated the analyses when either shifting the length or timing of the hazard and control periods relative to case onset; classifying exposures to wells based on annuli (ie, rings) from the place of residence; examining exposures to wells at further distances from the place of residence (ie, 6–10 km); considering only exposures to wells for which the place of residence was southeast of the well in preproduction (using the prevailing northwesterly winds in this region as a coarse proxy for the case being downwind of the well); or using only cases reported pre 2019 or post 2019 (appendix pp 2–3, 17).

We examined whether the association between preproduction wells and coccidioidomycosis incidence was modified by season, prevailing wind directionality, or patients' sociodemographic variables including age, race or ethnicity (self-reported), and sex, by including multiplicative interaction terms between exposure and each potential modifier in conditional logistic regression models (appendix p 3). For wind directionality, we compared downwind exposures with non-downwind exposures using the prevailing northwesterly winds as a coarse proxy. For each potential modifier, we compared models with and without the interaction term using likelihood ratio tests (ie, analysis of deviance with a χ^2 test). We used a p value of <0.05 to indicate statistical significance. To minimise the number of statistical tests, we conducted our analyses for only the binary and continuous exposure models at the 0–5-km distance bin. We present stratified effect estimates for descriptive purposes.

To estimate the proportion of the population exposed to newly constructed wells within 5 km of their places of residence during the study period, we used population estimates for Kern County from CA-POP, a publicly available, high-resolution dataset of gridded population estimates, which provides population estimates in 100-m grids based on downscaled census block data (figure 2).²² Using this population estimate, we calculated the proportion of the study population (1) exposed to at least one well in preproduction during the study period and (2) one to two preproduction wells, three to seven preproduction wells, eight to 22 preproduction wells, or 23 or more preproduction wells within a single 90-day window in the study period, corresponding to quartiles 1–4. We then identified the highest quartile exposure across 90-day sliding windows of time for each 100-m grid cell (ie, each grid cell fell into a single quartile to avoid double counting). All statistical analyses were conducted in R (version 4.3.2).

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, writing of the manuscript, or the decision to submit for publication.

Results

A total of 29 071 coccidioidomycosis cases were reported in Kern County between 2007 and 2022 (figure 1). During this period, we identified 26 360 newly constructed oil and gas

wells (ie, wells in preproduction). The annual numbers of incident cases and preproduction wells were weakly correlated (Pearson's $r=0.063$). Restricting our analysis to only cases occurring within 5 km of a well in preproduction during the study period resulted in a total of 23 220 cases and 13 890 wells (table; figure 2). 6917 (29.8%) of 23 220 cases had a preproduction well within 5 km of their place of residence during their hazard periods (ie, 49–139 days before the estimated date of case onset; appendix p 10). The number of preproduction wells within 5 km of a place of residence during the hazard period varied widely by case, ranging from no wells to 177 wells (appendix p 10). The timing of oil and gas development was evenly distributed across seasons (3412 [24.6%] of 13 890 wells had a spud date in winter; 3721 [26.8%] in spring; 3414 [24.6%] in summer; and 3343 [24.1%] in autumn).

We estimated that 658 108 (72.4%) of 909 282 Kern County residents lived within 5 km of at least one well that was in preproduction between 2007 and 2022 (appendix p 11). Furthermore, approximately 116 020 (12.8%) of 909 282 county residents were exposed to 23 or more preproduction wells at this distance within a single 90-day period in this time.

We found significant positive associations between exposure to preproduction wells and odds of coccidioidomycosis (figure 4). Under the binary exposure model, the odds of coccidioidomycosis were 12.5% higher (95% CI 5.8–19.6) when there was at least one well in preproduction within 5 km of the patient residence during the hazard period compared with those when there were no wells (figure 4A). Under the continuous exposure model, the odds of infection increased by 0.7% (CI 0.4–0.9) for each additional preproduction well within 5 km of the individual's place of residence (figure 4B). Similarly, under the quartile exposure model, the odds of coccidioidomycosis incidence were 19.4% higher (5.7–35.0) with exposure to one to two wells (first quartile) and 37.7% higher (19.4–58.9) with exposure to at least 23 wells compared with those when there was no exposure to wells (fourth quartile; figure 4C).

We observed consistent positive associations between exposure to preproduction wells and risk of coccidioidomycosis for all distance bins. For both binary and continuous exposures, effect estimates were generally larger for exposures at closer distances than for those at further distances. When considering continuous exposures in annuli rather than buffers, the associations were largest for the 0–1-km annulus (appendix p 23). However, associations at the 0–1-km distance were not statistically significant under the binary exposure classification (figure 4A). We might not have had sufficient power to detect associations at this distance as fewer than 1% of individuals with reported coccidioidomycosis were exposed to a preproduction well within 1 km of their place of residence during the hazard period (appendix p 9).

For our main model, the odds ratios for the negative control exposures were typically non-significant, indicating

	Individuals with coccidioidomycosis living within 5 km of a preproduction well (N=23 220)	All Kern County residents (%) (N=909 282)	All California residents (N=38 443 952)
Age, years			
0–17	2568 (11.1%)	267 329 (29.4%)	9 111 217 (23.7%)
18–64	18 089 (77.9%)	550 116 (60.5%)	24 296 578 (63.2%)
≥65	2520 (10.9%)	91 837 (10.1%)	5 036 158 (13.1%)
Race or ethnicity			
Asian or Pacific Islander	427 (3.1%)	40 918 (4.5%)	5 420 597 (14.1%)
Black	864 (6.4%)	49 101 (5.4%)	2 152 861 (5.6%)
Hispanic	6781 (50.0%)	470 099 (51.7%)	14 800 922 (38.5%)
Non-Hispanic White	4583 (33.8%)	325 523 (35.8%)	14 685 590 (38.2%)
Sex			
Female	10 455 (45.0%)	441 911 (48.6%)	19 298 864 (50.2%)
Male	12 715 (54.8%)	467 371 (51.4%)	19 145 088 (49.8%)
Season of onset			
Autumn (September–November)	7693 (33.1%)	..	..
Spring (March–May)	4406 (19.0%)	..	..
Summer (June–August)	5588 (26.1%)	..	..
Winter (December–February)	5538 (23.9%)	..	..

The study population consisted of all coccidioidomycosis cases reported in Kern County, CA, USA, that occurred within 5 km of a preproduction well during the study period (2007–22). Sociodemographic characteristics for all Kern County and state residents were drawn from the US Census Bureau's American Community Survey and reflect annual averages across 2007–22 to facilitate comparisons with the study population. Race or ethnicity percentages are based on the number of cases with reported race or ethnicity (n=13 562). Not all possible race or ethnicity categories are listed here so percentages do not sum to 100.

Table: Characteristics of the study population compared with Kern County and California residents

no statistically significant residual confounding, except for the 0–4-km and 0–5-km distances under the binary exposure model (appendix p 19). By including the negative control exposure in the model, the effect estimates for the primary exposure were generally lower, as some of the association between preproduction well exposure and infection risk due to residual confounding might have now been absorbed by the negative control (appendix pp 6, 13).

Our results were robust to alternative specifications of the hazard period (ie, 21–111 days, 35–125 days, 42–132 days, and 56–146 days before case onset as well as the 60-day exposure window) and exposure classification (appendix pp 20–23). Effect estimates remained statistically significant but were attenuated for exposures at greater distances (eg, 0–6 km to 0–10 km; appendix p 24). Results were broadly similar across all distances and exposure metrics when restricting exposures to wells for which the place of residence in question was to the southeast (ie, downwind given the prevailing wind direction in this region; appendix p 25) or when considering only cases that were reported pre 2019 or post 2019 (appendix p 26).

We observed evidence of effect modification by patient age and race or ethnicity but not by sex (appendix pp 5, 14, 28–30). In particular, the two age categories 0–17 years and 18–64 years typically had higher effect estimates for associations between well exposure within 0–5 km (in the binary model) and odds of coccidioidomycosis incidence than that observed for the 65 years or older age category. Individuals

For more on the Survey, see <https://www.census.gov/data/developers/data-sets/acs-1year.html>

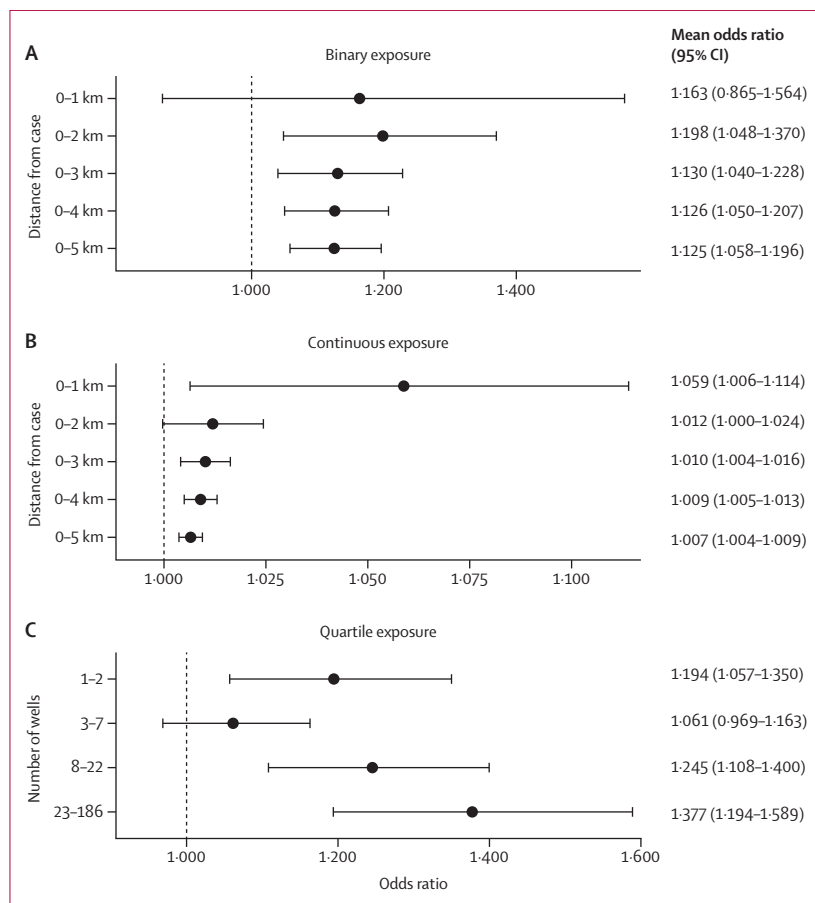


Figure 4: Odds ratios representing the odds of coccidioidomycosis, considering exposure to oil and gas well construction at varying distances from the place of residence

The mean and 95% confidence intervals for the odds ratios are shown for the binary (A), continuous (B), and quartile (C) exposure models, and the corresponding values are listed in the right panel. For the quartile exposure, we focused on exposures occurring between 0 km and 5 km and considered only non-zero exposures when calculating quartiles, with the reference group being no exposure to a well in preproduction. The corresponding odds ratios under alternative specifications are presented in the appendix (pp 20–22).

in the Hispanic and non-Hispanic White race or ethnicity categories had slightly higher effect estimates at the 0–5-km distance in the binary and continuous models, but this pattern did not appear consistent across other distance bins (appendix pp 5, 14, 30). We did not observe evidence of effect modification by whether a place of residence was southeast of a well, except in the models examining continuous exposure to wells within 0–5 km from the place of residence. In this circumstance, the effect estimate was lower for individuals who resided southeast of a well than for those who did not (appendix pp 14, 31).

Formal interaction tests did not support effect modification by season. However, when we stratified the analyses by the season of case onset, we found that the number of preproduction wells was significantly associated with increased odds of coccidioidomycosis in the autumn and winter, but not in the spring or summer (figure 5; appendix p 27). Specifically, for each preproduction well within 5 km of a place of residence, the odds of infection

increased by 1.3% in both autumn (95% CI 0.7–1.9) and winter (0.8–1.9). The lag between the exposure window and case onset corresponded to wells in preproduction during the summer and autumn.

Discussion

We found that the construction of new oil and gas wells was significantly associated with increased risk of coccidioidomycosis in a highly endemic region (Kern County, CA, USA). To our knowledge, although our findings represent a novel source of exposure risk, they are consistent with previous evidence for the role of soil disturbance and dust generation in coccidioidomycosis infection. Because *Coccidioides* spp reside in soil, processes that disturb soil and generate dust can facilitate the aerosolisation and dispersal of infectious spores.²³ Oil and gas development involves several major soil-disturbing processes,^{9,11,12} and the initial land disturbance has been estimated to extend across 2.5–5 acres.²⁴ Similarly, previous work has found significantly elevated PM_{2.5} concentrations near newly constructed wells,¹⁰ and fine mineral dust concentrations within PM_{2.5} have also been strongly associated with coccidioidomycosis incidence in California.⁸ Together, these findings support a mechanistic pathway linking well construction, dust generation, and coccidioidomycosis infection risk. Our results also extend previous evidence of the health consequences of oil and gas development beyond adverse birth outcomes, respiratory disease, cancer, and all-cause mortality¹³ to include transmission of an environmentally mediated infectious disease.

We observed consistent positive associations between proximity to preproduction wells and odds of coccidioidomycosis (figure 4). These associations were established when rigorously controlling for confounding factors with minimal or no variation over time (eg, occupation, socioeconomic status, proximity to a health-care facility, or propensity to seek care).^{6,19} Associations were generally stronger at smaller distances than at greater ones, although fewer than 1% of cases had any preproduction wells within 1 km of the individual's place of residence before disease onset (appendix p 10). Conversely, exposure to preproduction wells at greater distances was common: 658 108 (72.4%) of 909 282 of Kern County residents had a preproduction well within 5 km of their place of residence during the study period (ie, 2007–22) and 116 020 (12.8%) had 23 or more such wells in a single 3-month period (appendix p 11). Together, these findings suggest that particulate matter emissions associated with well construction might not be confined to the immediate well site but could contribute to infection risk in the surrounding communities. The high level of exposure to oil and gas well construction among this population and increased risk of infection given the exposure suggest a potentially large population-level impact on coccidioidomycosis incidence in this region. Population exposure might have decreased in recent years compared with previous years as the

number of newly constructed wells was highest during the 2010–20 period and declined thereafter.

We found that the positive association between oil and gas well construction and coccidioidomycosis was significant for case onset in autumn or winter in stratified analyses but not in spring or summer (figure 5). Given the lag between infection and case reporting, this result most closely reflects higher risk from exposure to wells in preproduction during summer and autumn. The finding aligns with previous evidence that exposure to dust during the summer and autumn months poses a higher risk of coccidioidomycosis than that observed in other seasons, putatively due to hot and dry conditions promoting greater spore lysis and airborne dispersal.^{25,26}

Although several previous studies have found associations between coccidioidomycosis severity and older age, male sex, and ancestral heritage (namely Black race and Filipino ethnicity),^{6,27} we did not observe consistent patterns that align with these findings (appendix pp 28–30). Specifically, although interaction tests indicated that associations varied significantly by age and race or ethnicity, our effect estimates were typically higher for younger age categories and were inconsistent for race or ethnicity categories across both models and distance bins. These patterns could reflect differences in opportunities for exposure—such as occupational or recreational activities, or time spent outdoors—and thus should be interpreted cautiously. For example, previous associations between male sex and hospitalisation for coccidioidomycosis²⁸ might reflect greater occupational exposure to *Coccidioides* spp spores (eg, through construction or agricultural work).^{29,30} Because our exposure metric is based on residential proximity to wells, it does not capture individual-level occupational or behavioural exposure pathways, nor does it assess variation in pathogen concentration or case severity. Thus, additional work is needed to distinguish whether and how demographic characteristics can be risk factors for *Coccidioides* exposure versus disease outcomes.

Our study has several limitations, including the potential for exposure misclassification. We assessed exposure on the basis of reported residence at the time of diagnosis, which might not reflect the true location of exposure to *Coccidioides* spores (eg, workplace exposure). Similarly, the incubation period, time to health-care seeking behaviour, and diagnostic testing could vary among individuals, motivating the 3-month exposure period (ie, 1.5–4.5 months before case onset) and sensitivity analyses applying different lags between exposure and case onset. However, if the times between exposure and case onset were substantially different, our effect estimates might also be subject to exposure misclassification. In both instances, the potential exposure misclassification would likely be non-differential between the hazard and control periods, thereby biasing our associations towards the null. Furthermore, our sensitivity analyses confirmed that our results were robust to alternative specifications of lags between exposure and case onset.

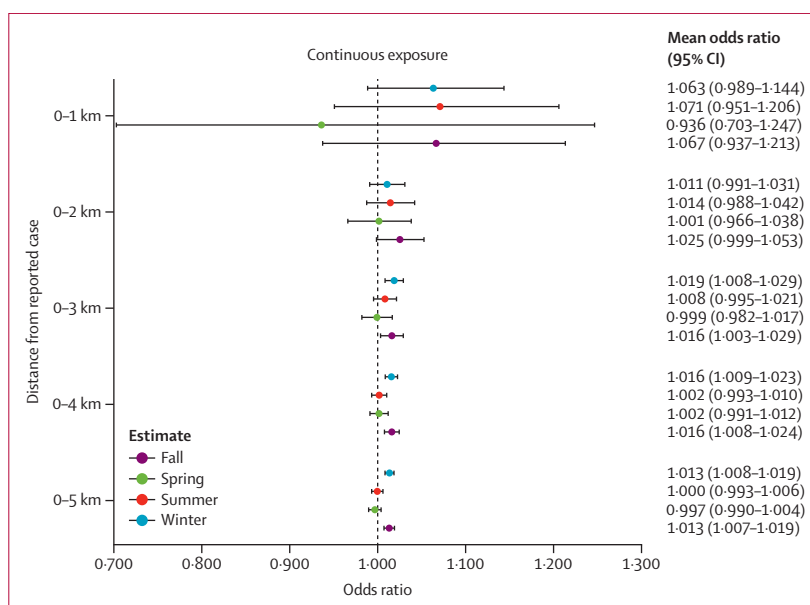


Figure 5: Odds ratios representing the odds of coccidioidomycosis incidence, considering exposure to oil and gas well construction at varying distances from places of residence, stratified by season of case onset. Odds ratios are shown for only the continuous exposure model. The season-stratified analyses under the binary exposure model are provided in the appendix (p 27).

We only had data on reported cases and therefore did not capture individuals with coccidioidomycosis who did not experience symptoms, did not seek medical care, or were misdiagnosed. This constraint might have reduced our power to detect associations between preproduction wells and infection risk and could have biased our associations towards the null if individuals living near oil and gas wells and other hazards were less likely to seek medical care. Furthermore, we note that our results reflected the average effect of exposures to preproduction wells across the study region, but these effects might vary spatially, with elevated risk associated with exposure to some wells and little to no risk associated with others. True exposure risk is probably higher at smaller distances (given gravitational settling of spores) and therefore non-uniform within a given exposure bin (eg, 0–1 km). However, the low case counts precluded robust estimation of associations at finer spatial resolutions.

Our consideration of what constituted downwind exposures was simply based on prevailing wind directionality in this region, given a scarcity of finely resolved (eg, sub-daily <10 km), publicly available wind data. Thus, the counter-intuitive finding, in which effect estimates were lower for individuals with places of residence downwind of preproduction than for those not living downwind might reflect this coarse consideration, as strong wind events countering the prevailing winds could have occurred and dispersed spores to residences classified as non-downwind.

Oil and gas well activity is one of several potential sources of soil disturbance and dust exposure in this region and should be interpreted in the context of a broader disturbance landscape that includes occupational, recreational,

and environmental sources of exposure. Although the specific sources of exposure risk are not well characterised, they might include a wide range of activities, such as construction, extraction, mining, excavation of undisturbed land that is unrelated to oil and gas drilling, agricultural work or other digging and landscaping, archaeological work, off-road vehicle use, and wildland firefighting,^{31,32} as well as natural disasters or extreme weather outbreaks events, such as earthquakes, wildfires, and dust events.^{33–35} Coccidioidomycosis outbreaks in Kern County have involved construction workers from a culvert expansion project in 2008,³⁶ wildland firefighters in 2021,³² and attendees of an outdoor music festival in 2024.³⁷ Furthermore, fugitive dust emissions in the Central Valley of California were shown to have increased between 2008 and 2022 in association with an expansion of fallowed land.³⁸ These potential sources of exposure are not mutually exclusive with exposure to oil and gas well construction, meaning that this study population could have been exposed to *Coccidioides* via multiple, overlapping sources. However, for other dust-generating activities to confound our results, the activities would need to be both temporally and spatially correlated with preproduction well activity within the specified hazard and control periods. We believe that other dust-generating activities are unlikely to fully explain the observed associations, particularly given the minimal correspondence in temporal trends in preproduction well activity and employment in construction-related and agriculture-related activities—a coarse proxy for broader earth-moving activity (appendix p 32).

Given previous findings on the seasonality of coccidioidomycosis and the results from our season-stratified analyses, shifting soil-disrupting activities away from summer and autumn months could potentially help to reduce coccidioidomycosis risk. For example, as the majority of precipitation in California occurs in the winter, earth-moving activities occurring in that season might pose lower risk due to reduced dust generation, and potentially reduced aerosolisation of *Coccidioides* spores, since aerosolisation is thought to be favoured in hot, dry conditions.²⁶ Simultaneously, measures including wetting the soil before disturbance to reduce dust, avoiding dusty areas when possible, and, when unavoidable, keeping windows closed, wearing a properly fitting N95 respirator (or doing both) can collectively help to reduce infection risk.³⁹ People living in endemic regions near sites where major soil disturbance activities are planned should be made aware of the potential risk to ensure they develop personal prevention measures and seek care and undergo possible coccidioidomycosis testing in a timely manner if compatible symptoms develop. As no vaccines are currently available for coccidioidomycosis and therapeutic options are few, measures aimed at reducing environmental exposure to *Coccidioides* spores are crucial to protecting public health.

Additional research is needed to clarify how risk factors for *Coccidioides* exposure (eg, soil-disturbing occupations or recreational activities and proximity to dust-generating

events) interact and the extent to which intervention would reduce risks. Further work is also needed to understand the mechanistic links between the construction of oil and gas wells and infection, including identifying the types of activities most likely to aerosolise spores, and the role of wind in spore dispersal, including the wind speeds necessary for aerosolisation and the distances over which aerosolised spores can travel.

Contributors

JRH, DJXG, and SKC originally conceived this research project. JRH, LIC, SKC, AKW, and JVR designed and conducted the analysis, with conceptual guidance from DJXG, GLSC, DV, SJ, JT, JB, and EAE. JRH and LIC wrote the original draft of the manuscript. All authors had full access to the data and reviewed and edited the final draft and approved submission. LIC and JRH directly accessed and verified the underlying data. LIC had final responsibility for the decision to submit the manuscript for publication.

Declaration of interests

JB is a physician member of the Johns Hopkins Clean Air Data Safety Monitoring Board and California Air Resources Board and has received funding unrelated to the present research from the California Energy Commission. All other authors declare no competing interests.

Data sharing

The R script used to conduct the data analysis is available in a publicly available GitHub repository. Human case data are protected health information and can only be obtained by submitting a formal data use request to the California Department of Public Health and Infectious Disease Branch, Surveillance and Statistics Section. Oil and gas data are publicly available from the California Geologic Energy Management Division, USA.

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